

Degradable hyperbranched polymer with fouling resistance for antifouling coatings

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ABSTRACT

The controlled release of antifoulant is critical in marine antifouling, which is profoundly influenced by the compatibility between polymer resin and antifoulant. Conjugation of polymers with active molecules provides a promising strategy to solve the problem. In this study, we report the synthesis of hyperbranched polymers containing antifouling main-chains and degradable branching points by using reversible addition-fragmentation chain-transfer (RAFT) polymerization. The copolymerization of butyl methacrylate, vinyl-functional E-cone and divinyl-functional poly(ϵ -caprolactone) (PCL) yields a degradable hyperbranched antifouling polymer (DHAP). As revealed by ^1H NMR, the vinyl conversion linearly increases over reaction time during RAFT polymerization with a final conversion rate above 90 % after 24 h. The branching degree of DHAP increases with the content of PCL segment. Quartz crystal microbalance with dissipation (QCM-D) measurements show the degradation of DHAP is controlled. Antibacterial assays indicate that DHAP coating has good antibacterial activity. Thus, DHAP represents a promising system for antifouling coatings.

1. Introduction

Marine biofouling formed by the settlement of undesirable marine organisms on artificial surfaces immersed in seawater is a serious problem for maritime industries [1,2]. The use of coatings containing various antifoulants is established as the most economical, convenient, and effective strategy. Tributyltin self-polishing copolymer based coatings are the most effective coatings against marine biofouling, but they were banned in 2008 because of their high toxicity to nontarget organisms [3,4]. Recently, many environmentally friendly antifouling systems have been reported, such as tin-free self-polishing coatings [5,6], fouling release coatings [7,8], amphiphilic polymers [9–11], zwitterionic polymers [12,13], biodegradable polymers [14–16] and antibacterial polymers [17–20]. Moreover, antifoulants from natural products [21,22] and synthetic compounds [23] were developed. Antifoulants are generally physically mixed into polymer matrices to inhibit the adhesion and growth of biofouling [24,25]. The antifouling performance of coating containing antifoulants depends largely on the antifoulant release rate and loading dose. Actually, the antifoulant release is difficult to control since the antifoulants often are not well compatible with polymer matrices, this is, they easily migrate and aggregate in the polymer matrices [26].

Recently, polymeric prodrugs or polymers conjugated with active molecules were reported [27–29]. These macromolecular prodrugs exhibit advantages over low-molecular-weight precursors, such as increased compatibility, controllable release, and improved pharmaceutical effects [30–32]. In such polymeric prodrugs systems, drugs are released by the cleavage of linked groups [33]. Likewise, antifoulant can be conjugated with a polymer making its controlled release possible. For examples, pyridine-diphenylborane [34], glyphosate [35], benzoisothiazoline ketone [20], etc., have been chemically bonded to polymers to realize controlled release. However, the release of such antibacterial polymer with non-degradable main chain largely depends on the hydrolysis of side groups, which is determined by water shear. To address this problem, degradable antifouling polymers which can form a self-renewing surface to improve antifouling performance have been developed [18]. With degradable main chain and hydrolysable antifoulant groups, a degradable antifouling polymer may realize the controlled release of antifoulants. Yet, such a polymer is linear and generally has high viscosity. Hyperbranched polymers usually exhibit high solubility, low melt viscosity, low crystallinity, and high end-up functional group density compared with linear polymers [36–40]. Furthermore, degradable hyperbranched polymers can completely degrade into small fragments owing to their short branched chains.

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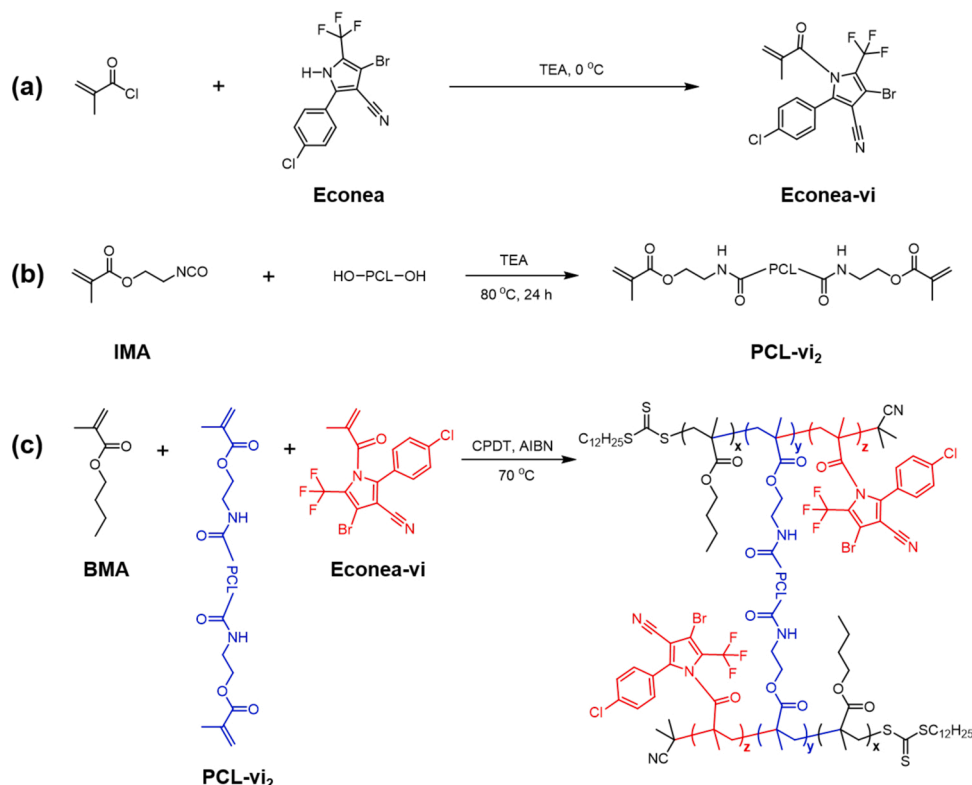
In the present study, we prepared a degradable hyperbranched antifouling polymer (DHAP) via reversible addition-fragmentation chain-transfer (RAFT) polymerization of butyl methacrylate (BMA), vinyl-functional Ecomea (Ecomea-vi) and divinyl-functional poly(ϵ -caprolactone) (PCL-vi₂) (Scheme 1), where the BMA is to improve the film-forming property, the Ecomea confers fouling resistance, and PCL segments allow the polymer to degrade. The degradable branch point facilitates decomposition of the DHAP into small molecules instead of particles without the so-called microplastic pollution. Moreover, the presence of branched structure can reasonably reduce the crystallinity of PCL, further improving the degradation rate. Herein, we focus on the synthesis of DHAP and preliminary evaluation of antifouling and degradation properties. Our goal is to develop an environmentally friendly and effective antifouling coating.

2. Result and discussion

The conversion of DHAP was examined by proton nuclear magnetic resonance spectroscopy (¹H NMR). A high concentration of PCL-vi₂ (33 mol % with respect to the total vinyl monomer feed) was used to prepare a polymer with multiple branching. Fig. 1a shows the time dependence of the vinyl conversion during RAFT polymerization. The vinyl conversion rate is linear over the reaction time, indicating that chain growth is controlled [41]. The conversion rate exceeds 90 % in 24 h, and the proton signals at 6.10 and 5.59 ppm assigned to the vinyl group almost disappear (Fig. 1b). The facts indicate that nearly all monomers convert into polymer. As shown in Fig. 1b, we can observe the peaks at 7.44 (c) and 7.32 (d) ppm attributed to the protons in the benzene ring from the antifoulant Ecomea. Peaks at 4.06 (i), 2.31 (h), 1.64 (f) and 1.38 (g) ppm are all assigned to caprolactone (CL) units from PCL-vi₂. The hyperbranched copolymer composition was determined by integrating peaks a, c, and e. A 25/5/1 feed ratio of BMA/Ecomea-vi/PCL-vi₂ produced a copolymer with 16 wt% of Ecomea and 34 wt% of CL. By calculating the molar ratio of CL content, we know the branching degree (BD) of the above polymer is 3.7 as shown in Table 1.

Since the antifoulant content generally determines the antifouling ability, we prepared DHAP with different Ecomea contents (0, 2, 7, and 16 wt%) by changing the molar ratio of BMA and Ecomea-vi. The characterization data for the resulting polymers P1 – P4 are listed in Table 1. The antibacterial properties of the DHAP coatings were investigated via the plate count method. As shown in Fig. 2, there were a large number of bacterial colonies covering the plate of the sample without Ecomea (Evi₀, P1). In contrast, relatively few bacterial colonies were observed on the plate of DHAP with 2 wt% Ecomea (Evi_{0.02}, P2), indicating that Ecomea has antibacterial activity even at low content. As Ecomea content increases, the number of bacterial colonies decreases, indicating that DHAP with Ecomea groups has an excellent antibacterial property. As reported before, the settlement of bacteria on the immersed surfaces is a crucial process in the formation of marine biofilms [42–44]. Accordingly, we studied the ability of the polymers P1 – P4 to resist biofilm formation of the typical marine bacteria *Pseudomonas* sp. (Fig. S3). As Ecomea content increases, the biofilm coverage decreases, and the absorbance value of crystal violet decreases. The facts indicate that the Ecomea group plays a critical role in preventing biofilm formation. Thus, the polymer conjugation with Ecomea is effective to inhibit marine biofouling probably attributed to the contact killing and the release of Ecomea after degradation.

Antibacterial polymers usually have a limited antifouling ability in complex marine environments since dead bacteria can still cover the surface and may cause the antibacterial polymer to lose its function [45, 46]. As discussed above, PCL-based polymers can degrade into small molecules dispersing into seawater, forming a constantly self-renewing surface to inhibit the adhesion of marine organism [47]. The renewal rate depends on the degradation rate of the polymer. Therefore, we adjusted PCL content for RAFT polymerization to synthesize DHAP polymers with different CL contents (0, 10, 18, and 34 wt%). As shown in Table 1, as PCL content increases from 0 to 7.5 mol%, the CL content of DHAP increases from 0 to 34 wt% and BD increases from 0 to 3.7. These parameters indicate that PCL-based hyperbranched polymers have already been prepared.



Scheme 1. Synthesis of vinyl-functional Ecomea (a), divinyl-functional poly(ϵ -caprolactone) (b), and degradable hyperbranched antifouling polymer (c).

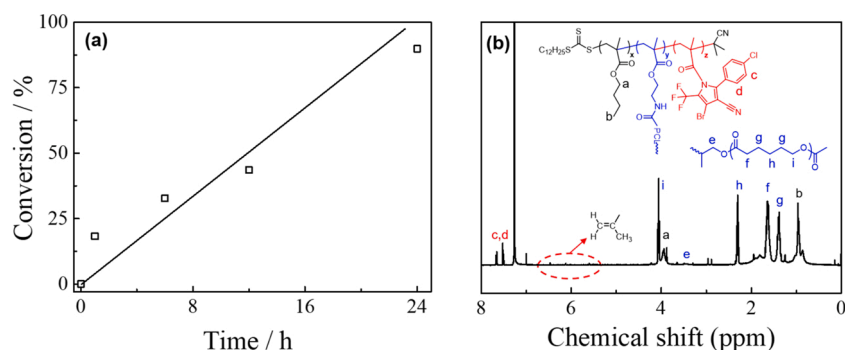


Fig. 1. (a) Time dependence of vinyl conversion for DHAP with a molar ratio of BMA/Evi/PCL- vi_2 of 25/5/1 and (b) ^1H NMR spectra of DHAP after 24 h reaction.

Table 1

Characterization data of hyperbranched polymers.

Polymer	BMA/Evi ^a /X ^b /Y ^c	Reaction time(h)	Evi (wt%) ^d	CL (wt%) ^d	$M_n \times 10^{-4}$ (g/mol) ^e	$M_w \times 10^{-4}$ (g/mol) ^e	D_m ^f	BD (mol%)	$M_n^* \times 10^{-4}$ (g/mol) ^f
P1	30/0/1/0.15	24	0	32	2.59	11.70	4.50	2.8	0.25
P2	29.5/0.5/1/0.15	24	2	32	2.48	13.00	5.24	2.9	0.23
P3	28/2/1/0.15	24	7	33	1.83	9.75	5.33	3.1	0.22
P4	25/5/1/0.15	24	16	34	1.48	8.40	5.70	3.7	0.21
P5 ^g	25/5/0/0.15	24	18	0	1.42	2.80	1.97	0	1.22
P6	25/5/0.01/0.15	24	16	10	1.39	5.32	3.83	0.8	0.31
P7	25/5/0.25/0.15	24	15	18	1.37	7.38	5.37	1.5	0.28

^a Vinyl-functionalized Econe.

^b Divinyl-functionalized PCL.

^c molar ratio = $[\text{CPDT}]_0/[\text{AIBN}]_0 = 2/1$.

^d Determined by ^1H NMR.

^e Obtained by gel permeation chromatography. BD = branching degree.

^f Number average molecular weight after accelerated degradation for 24 h.

^g Linear polymer.

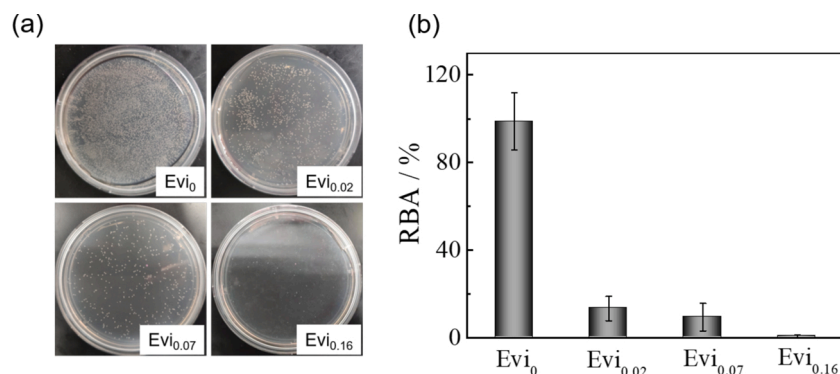


Fig. 2. (a) Photographs of marine bacterial colonies of *Pseudomonas* sp after incubation with treated coatings containing Econe (0, 2, 7, and 16 wt%) at 30 °C for 24 h and (b) the relative bacterial adsorption (RBA).

We investigated the enzymatic degradation of DHAP in artificial seawater (ASW) at 25 °C by quartz crystal microbalance with dissipation (QCM-D) to evaluate the self-renewing properties. Fig. 3a shows the time dependences of frequency shift (Δf) and dissipation shift (ΔD) for the enzymatic degradation of DHAP as a function of CL content. The frequency shift (Δf) generally increases with the decrease in mass on the sensor surface [48]. For $h\text{-PCL}_0$ (P5), Δf decreases after the introduction of lipase PS and then gradually levels off. There is still a notable decrease in Δf relative to the baseline after rinsing with ASW, indicating that the mass increases due to the absorption of lipase PS. For $h\text{-PCL}_{0.10}$ (P6), a small increase in Δf relative to the baseline indicates the contribution to mass from the degradation is more than that from absorption of lipase PS. For $h\text{-PCL}_{0.18}$ (P7) and $h\text{-PCL}_{0.34}$ (P4), we observe an obvious increase in Δf relative to the baseline, indicating that its mass loss is more than that of $h\text{-PCL}_{0.10}$ (P5). This is because more ester chains in the

polymer degrade into small molecules dispersing into the solution. Considering a further increasing CL content could lead to a higher crystallinity that lowers the degradation rate, we measured the crystallinity of the DHAPs as a function of CL content (Fig. S4). All the DHAPs with different CL contents have no melting enthalpy change, indicating that they have weak crystallinity in the range of 0–34 wt% CL contents. Clearly, the degradation rate of hyperbranched polymer is largely affected by the CL content. On the other hand, the dissipation (ΔD) generally increases with thickness but decreases with the rigidity of a layer [49]. The increase in ΔD relative to the baseline for $h\text{-PCL}_0$ (P5) indicates a viscoelastic layer formed by the adsorption of lipase PS. For $h\text{-PCL}_{0.10}$ (P6), the increase in ΔD is lower than $h\text{-PCL}_0$ (P5) because lipase PS leads to a thick layer but the degradation causes a thin layer and the latter is dominant. After rinsing with ASW, $h\text{-PCL}_{0.18}$ (P7) and $h\text{-PCL}_{0.34}$ (P4) exhibit a lower ΔD , indicating a lower thickness because

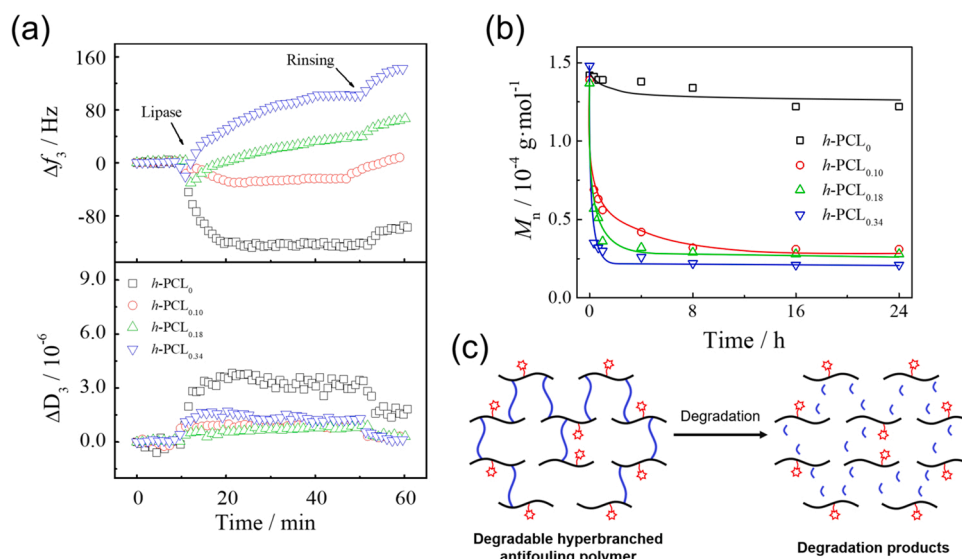


Fig. 3. (a) Time dependence of the frequency shift (Δf) and dissipation shift (ΔD) for the enzymatic degradation of DHAP with different CL content in ASW at 25 °C, (b) variation of molecular weight of DHAP degraded in potassium hydroxide methanol solution at 30 °C, and (c) schematic of degradation of DHAP.

of fast degradation. Thus, we can obtain a series of DHAPs with different degradation rate by changing CL content to apply to different occasion.

The size of the degradation fragments could affect the surface self-renewal rate and leads to negative impact on the marine environment, we also studied the accelerated hydrolytic degradation of the DHAPs as a function of time. Fig. 3b shows the variation of the number average molecular weight (M_n) of DHAP under accelerated conditions. For h -PCL₀ (P5) without ester bonds, M_n slightly decreases from 14,000 to 12,000 g/mol after accelerated degradation, indicating its main chain hardly degraded. For h -PCL_{0.10} (P6), h -PCL_{0.18} (P7) and h -PCL_{0.34} (P4), M_n quickly decreases and gradually levels off because of the breaking of ester bonds, and the final products have M_n of 2000 – 3000 g/mol, indicating that DHAP degrades into small fragments. After degradation for 24 h, the molecular weight of the final degradation product decreases slightly with CL content (Fig. S5). This is probably because higher divinyl-functionalized PCL content leads to a higher BD of the hyperbranched polymer (Table 1), which would break into a larger number of small fragments after degradation. Anyhow, DHAP is shown to degrade into small fragments so that the microplastic pollution is avoided protecting the marine ecological environment.

3. Conclusion

We have prepared a degradable hyperbranched antifouling polymer (DHAP) consisting of Econe groups and PCL segments by RAFT polymerization. The conversion of the vinyl group is linear over the reaction time in the polymerization with final conversion rate exceeding 90 % after 24 h. The branching degree (BD) of DHAP increases with PCL segment content. Such hyperbranched degradable polymers conjugated with Econe exhibit excellent antibacterial and anti-biofilm activities. By adjusting PCL segment content, the degradation rate of DHAP can be well regulated. All DHAP samples can completely degrade into small fragments to prevent environmental pollution. With an antifouling and degradable surface, DHAP is expected to be used in marine antifouling.

Data availability

The raw/processed data required to reproduce these findings cannot be shared at this time as the data also forms part of an ongoing study.

CRediT authorship contribution statement

Xiaoqing Ai: Conceptualization, Visualization, Writing - original draft. **Liqin Mei:** Writing - review & editing. **Chunfeng Ma:** Supervision, Resources, Writing - review & editing. **Guangzhao Zhang:** Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.porgcoat.2021.106141>.

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